

Table 1. [All-in-one] Quantitative comparison on PLAMD under the all-in-one setting. Panels (a)–(g) report restoration performance on seven degradation types (blur, haze, inpainting, low-light, noise, rain, snow); panel (h) reports their macro-average; panel (i) evaluates source preservation with an independent frozen verifier. Best and second-best results are in red and blue, respectively.

(a) Deblurring				(b) Dehazing				(c) Inpainting			
Method	PSNR ↑	SSIM ↑	LPIPS ↓	Method	PSNR ↑	SSIM ↑	LPIPS ↓	Method	PSNR ↑	SSIM ↑	LPIPS ↓
AirNet	27.0546	0.9428	0.1901	AirNet	20.2208	0.8163	0.1322	AirNet	29.2976	0.9651	0.1077
DFPIR	28.2745	0.9511	0.1874	DFPIR	22.7988	0.8733	0.0789	DFPIR	31.3313	0.9750	0.0862
DehazeFormer	28.7632	0.9545	0.1640	DehazeFormer	23.3833	0.8938	0.0778	DehazeFormer	32.3211	0.9786	0.0628
Restormer	28.5392	0.9527	0.1708	Restormer	23.9411	0.9041	0.0689	Restormer	31.5548	0.9758	0.0817
PromptIR	28.3911	0.9519	0.1720	PromptIR	23.9489	0.9029	0.0665	PromptIR	31.5578	0.9759	0.0814
FFANet	27.5854	0.9463	0.1922	FFANet	21.9422	0.8522	0.0989	FFANet	28.2076	0.9590	0.1293
TransWeather	26.6901	0.9436	0.2303	TransWeather	19.4481	0.7956	0.1871	TransWeather	25.7172	0.9207	0.2674
Uformer	25.9079	0.9252	0.2093	Uformer	22.4933	0.8701	0.1121	Uformer	29.9937	0.9683	0.1135
SiblingRestore (Ours)	28.7663	0.9542	0.1704	SiblingRestore (Ours)	24.4303	0.9071	0.0607	SiblingRestore (Ours)	31.0613	0.9738	0.0914

(d) Low-light Enhancement				(e) Denoising				(f) Deraining			
Method	PSNR ↑	SSIM ↑	LPIPS ↓	Method	PSNR ↑	SSIM ↑	LPIPS ↓	Method	PSNR ↑	SSIM ↑	LPIPS ↓
AirNet	18.9177	0.7398	0.2509	AirNet	27.5903	0.9423	0.2085	AirNet	28.9739	0.9555	0.1535
DFPIR	24.8115	0.9543	0.1707	DFPIR	27.7502	0.9433	0.1852	DFPIR	32.0177	0.9717	0.1309
DehazeFormer	28.4432	0.9706	0.1223	DehazeFormer	28.0278	0.9462	0.1318	DehazeFormer	32.3790	0.9748	0.0976
Restormer	26.3311	0.9654	0.1335	Restormer	27.9777	0.9454	0.1578	Restormer	33.1685	0.9784	0.1071
PromptIR	23.5944	0.9476	0.1492	PromptIR	27.7880	0.9439	0.1681	PromptIR	32.8370	0.9755	0.1199
FFANet	28.8351	0.9699	0.1933	FFANet	27.4972	0.9414	0.2218	FFANet	30.0487	0.9606	0.1778
TransWeather	21.9976	0.8812	0.2438	TransWeather	25.0303	0.9084	0.2826	TransWeather	26.9519	0.9427	0.2170
Uformer	22.8807	0.8946	0.2691	Uformer	27.1845	0.9375	0.2260	Uformer	27.9066	0.9489	0.1720
SiblingRestore (Ours)	29.6594	0.9737	0.1335	SiblingRestore (Ours)	27.8104	0.9440	0.1590	SiblingRestore (Ours)	33.8350	0.9810	0.0903

(g) Desnowing				(h) Average Restoration Quality				(i) Source Preservation			
Method	PSNR ↑	SSIM ↑	LPIPS ↓	Method	PSNR ↑	SSIM ↑	LPIPS ↓	Method	Top-1 ↑ (%)	EER ↓ (%)	AUC ↑
AirNet	21.4407	0.7933	0.4020	AirNet	24.7851	0.8793	0.2064	AirNet	0.4442	0.1357	0.9395
DFPIR	23.9498	0.8573	0.3629	DFPIR	27.2762	0.9323	0.1717	DFPIR	0.5954	0.0998	0.9625
DehazeFormer	25.2842	0.8848	0.3066	DehazeFormer	28.3717	0.9433	0.1376	DehazeFormer	0.6708	0.0942	0.9692
Restormer	25.2195	0.8812	0.3276	Restormer	28.1046	0.9433	0.1496	Restormer	0.6562	0.0902	0.9702
PromptIR	24.5615	0.8687	0.3483	PromptIR	27.5255	0.9381	0.1579	PromptIR	0.6334	0.0944	0.9674
FFANet	22.1685	0.8148	0.4274	FFANet	26.6121	0.9206	0.2058	FFANet	0.5301	0.1082	0.9557
TransWeather	20.6996	0.7545	0.4932	TransWeather	23.7907	0.8781	0.2745	TransWeather	0.3181	0.1522	0.9254
Uformer	22.9318	0.8233	0.4049	Uformer	25.6141	0.9097	0.2153	Uformer	0.4883	0.1383	0.9366
SiblingRestore (Ours)	25.9708	0.8866	0.2981	SiblingRestore (Ours)	28.7905	0.9458	0.1433	SiblingRestore (Ours)	0.6981	0.0799	0.9749

Panel (h) macro-average = equal-weight mean of the seven per-degradation means (each degradation contributes equally; class sizes are equal on the test set). Panel (i): Top-1 = fraction of restored views whose clean-source is the top retrieval in the full 256-source gallery; EER/AUC are aggregated over the same per-view cosine scores against that gallery. Percent columns report % values. All numbers originate from official frozen-verifier evaluations on the same 256-source test split (all methods share one verifier and one LPIPS protocol; PSNR in dB, LPIPS AlexNet 256×256). Missing results would be shown as “-”; no method in this table lacks any degradation. Training protocols: all listed methods were trained on the same mixed seven-degradation dataset (all-in-one); siblingrestore (Ours, config A5 = deg+fixed-anchor) used sibling-group sampling and 20k steps, external baselines used independent sampling and 12k steps, so results are not claimed to come from a fully controlled common retraining budget. Single training seed (13) per method. Noise images have native sizes differing from clean; evaluation aligns clean to noise by Lanczos resize before scoring (project Method-B protocol).